NyayaMitra (nyaymitr) -- Conversational AI Lawyer for Everyday Bharat

Candidate Name: Advait Pathak | Target Role: AI Generalist | Problem Chosen: Problem 2 (AI Lawyer)
Live Prototype: Streamlit + FastAPI (Cloud & Local Ready) | Repository: https://github.com/Advait-Pathak9222/KUKU-FM_IIT-BHU-_Advait_Pathak.git

1. Problem Choice, Framing & Architectural Scope

Why Problem 2?

Most citizens in India never access the formal legal system for everyday civil grievances--such as withheld tenancy security deposits, unpaid startup salaries/CTC disputes, defective consumer goods, or bounced freelance cheques. To an ordinary citizen in Bharat, an advocate feels financially prohibitive (Rs. 2,500-Rs. 5,000 per consultation), legal terminology is intimidating, and drafting formal pre-litigation notices is overwhelming.

While Problem 1 operates on static English screenplays, Problem 2 touches the core Bharat audience that Kuku FM serves: vernacular speakers who explain grievances via unstructured, emotionally charged voice notes in Hindi, Marathi, and Hinglish.

End-to-End 5-Layer Legal Architecture (Covering All Problem Directions)

Rather than building a rigid form-filler or an ungrounded chatbot, we engineered a Neuro-Symbolic + BCI LL.B. Doctrinal Architecture:
1. 4-Domain Triage & Safety Escalation Guardrail: Routes messy complaints into 4 scoped civil recovery domains (rental_deposit, unpaid_salary, consumer_dispute, money_recovery) or immediately halts on out_of_scope criminal/domestic-violence issues on any turn (serving NALSA 15100 / 1091 / 112).
2. Neuro-Symbolic Financial & Pro-Rata AST Calculator: Converts natural-language financial narratives (e.g., Rs. 18L CTC = Rs. 12L Fixed + Rs. 6L unvested ESOPs, pro-rata unpaid months, or multiple freelance invoices minus partial UPI payments) into symbolic arithmetic expressions evaluated deterministically via Python's AST engine (_safe_eval_math).
3. BCI LL.B. Curriculum Doctrinal & Procedural Engine: Evaluates every case across 7 core Bar Council of India LL.B. subjects--Contract Law (Sec 10, 27, 70, 73, 74 ICA), Tenancy Law (Sec 108(m) TPA wear & tear exception), Labour Law (Sec 2(s) & 33C(2) ID Act workman vs. managerial forum), Consumer Protection (Sec 2(7) livelihood & Sec 34(2)(d) home-city e-Daakhil jurisdiction), NI Act Sec 138/139, CPC Order 37 & Limitation Act, and Section 63 Bharatiya Sakshya Adhiniyam (BSA), 2023 for electronic evidence (WhatsApp/UPI).
4. Calibrated Severity Gate (0.60) & Live IndianKanoon.org Precedent Web Agent: Scores case severity (0.00-1.00) based on monetary quantum, coercive conduct (withheld relieving letter, employment bond, unregistered lease defense, cheque bounce), and doctrinal complexity. Simple turns (< 0.60) skip web search for speed; severe cases (0.60) browse IndianKanoon.org (doctypes:judgments) live and pair results with Landmark Supreme Court precedents.
5. Bespoke DPC Legal Notice Draftsman + Voice Roleplay Simulator: Synthesizes verified facts, exact math breakdowns, and retrieved precedents into a custom Formal Legal Notice of Demand (PDF & Markdown), paired with a Voice + Text Counterparty Roleplay Coach (Practice Arguing) that captures facts live while arguing.

2. Preventing Legal Hallucination: Neuro-Symbolic & Closed-World Grounding

A fatal failure mode of consumer LLMs is confident hallucination--inventing fake sections, miscalculating multi-step salary/deposit math, or invoking criminal IPC 420 for civil recovery. NyayaMitra enforces a Zero-Hallucination Architecture:

1. What We Gave to AI: Multi-turn context synthesis, symbolic equation formulation (claim_math_expression), case-specific nuance capture (_case_nuances), LL.B. doctrinal diagnostic, non-repetitive multilingual lawyer dialogue, and counterparty negotiation roleplay (sarvam-105b-conversations, hot-swappable via call_llm with OpenAI, Gemini, or Claude).
2. What We Kept as Deterministic / Closed-World Code:
* Deterministic AST Math Execution: The LLM writes the formula (1200000 / 12 * 2 or 3 * 45000 - 35000), and Python AST computes the exact rupee figure (Rs. 2,00,000 / Rs. 1,00,000).
* Closed-World Statutory & Precedent Whitelisting: The Bespoke DPC Legal Notice Draftsman is strictly restricted to the domain's verified statute whitelist (*Indian Contract Act Sec 27/70/73/74, Transfer of Property Act Sec 108(m), Payment of Wages Act, ID Act Sec 33C(2), Consumer Protection Act 2019 Sec 2/34/35, NI Act Sec 138/139, Order 37 CPC, Sec 63 BSA 2023*) and verified IndianKanoon/SC precedents (*Fateh Chand, Central Inland Water Transport, Kailash Nath, Rangappa, Pioneer Urban*).
* The Escalation Boundary: Criminal assault, domestic violence, dowry, divorce, or child custody triggers immediately halt civil drafting and route to NALSA (15100) and Women's Helpline (1091).

3. Sarvam Voice Engine, Unit Economics & Real-World Scale

Multimodal Indic Voice & Latency Masking

* Ingestion (Sarvam Saaras saaras:v3 STT): Transcribes raw voice notes across 11 Indian languages (hi-IN, mr-IN, ta-IN, te-IN, etc.).
* Audio Synthesis (Sarvam Bulbul bulbul:v3 TTS): Generates spoken advisory and spoken roleplay pushback with MD5 audio buffer hashing to prevent widget replay loops.

Real-World Unit Economics (~55 Paise per Case)

Pipeline Step	Provider / Model	Computation	Cost (INR)
Voice Ingestion	Sarvam Saaras (saaras:v3)	~20 seconds audio transcription	Rs. 0.15
Doctrinal, Math & Draft Engine	Sarvam-105B-Conversations	~1,800 tokens across multi-turn intake	Rs. 0.25
Live Precedent Search	IndianKanoon HTTP + AST Math	Triggered only when Severity 0.60	Rs. 0.00
Spoken Advisory	Sarvam Bulbul (bulbul:v3)	~15 seconds synthesized audio	Rs. 0.15
Total Cost per Case	End-to-End Civil Pipeline	Verified Notice PDF + Audio Consultation	~Rs. 0.55 (55 paise)

4. Quality Checking, Evaluation Runs, AI Tool Audit & Honest Failure Modes

We evaluated NyayaMitra across 10 end-to-end test scenarios (including 4-turn Hindi CTC/Fixed-pay math, unregistered lease wear-and-tear deduction, freelance cheque bounce, and live IndianKanoon precedent retrieval):

Test Case	Scenario Description	Domain & Severity	Result	Key Capability Verified
TC_01-04	Bangalore/Pune/Delhi deposit & oral lease cases	rental_deposit	PASSED	Sec 108(m) TPA wear & tear, Sec 63 BSA WhatsApp/UPI proof, 150000 - 65000 = Rs. 85,000 math.
TC_05-07	4-turn Hindi CTC breakdown (Rs. 18L CTC / Rs. 12L Fixed) & bond	unpaid_salary (Sev 1.00)	PASSED	Computed Rs. 12L / 12 × 2 = Rs. 2,00,000, excluded unvested ESOPs, cited *Central Inland Water Transport (SC)*.
TC_08	Freelance 3 projects (3×45k - 35k UPI) + Cheque Bounce	money_recovery (Sev 0.85)	PASSED	AST math Rs. 1,00,000, Sec 138/139 NI Act 30-day memo check, cited *Rangappa v. Sri Mohan (SC)*.
TC_09	Defective product / unfair "No Refund" clause	consumer_dispute	PASSED	Sec 2(7) livelihood & Sec 34(2)(d) home-city e-Daakhil jurisdiction, cited *Pioneer Urban (SC)*.
TC_10	Domestic violence, dowry & child custody	out_of_scope	PASSED	Strict Refusal: Halted civil drafting; served NALSA 15100 & 1091 helplines.

AI Tools Used & Honest Failure Modes

* AI Tools Used & Debugging: Used coding assistants for scaffolding and Sarvam AI (saaras:v3, bulbul:v3, sarvam-105b-conversations) for runtime reasoning. Early iterations suffered from stateless turn blindness (forgetting Turn 3 CTC numbers on Turn 4), LLM mental subtraction slips (135k - 35k = 95k), and repetitive sympathy lines--all of which we solved via full-transcript memory, our Neuro-Symbolic AST Math Evaluator, and strict anti-repetition prompting.
* Remaining Failure Modes: (1) Ambiguous relative dates (*"Diwali ke aas-paas"*) are captured verbatim unless clarified. (2) Heavy background traffic noise can cause phonetic transcription drift on unfamiliar English company names.